

Revisiting the Eye-Level Effect

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The principle that “eye-level is buy level” underpins planogram optimization and retailer-manufacturer agreements. When a product is moved to eye-level, the remaining products on the same shelving unit can be reorganized in several different ways (i.e., planogram reorganization). This study examines how different planogram reorganizations affect the magnitude of the eye-level effect. Using data from a subset of a field experiment originally designed to test alternative planograms at a North American convenience store chain, we compare eye-level placement under two distinct planogram reorganizations. We find that planogram reorganization significantly moderates the eye-level effect. Moving products from stretch-level (stoop-level) to eye-level increased sales by 5.6% (10%) in one reorganization, but did not change sales in the other reorganization. We attribute this to partial visual search, where some customers scan only the stretch-level and eye-level shelves. Different planogram reorganizations create distinct choice sets for these customers, resulting in different levels of preference for products placed at eye-level. These differences also impacts overall sales, with the effective reorganization type varying by shelving unit. Retailers should therefore tailor planogram reorganizations based on product preference patterns within each shelving unit. Our counterfactual analysis reveals that compared to assuming no moderation effect of planogram reorganization, uniformly applying the dominant planogram reorganization strategy across all shelves can boost sales by 3.85%, while tailoring the reorganization for each shelving unit can increase sales by 4.32%.

Key words: Eye-level effect, planogram reorganization, retail operations, empirical research

1. Introduction

Customers rely on shelf layout to gauge product quality (Valenzuela and Raghuram 2015), product popularity (Valenzuela et al. 2013), and brand value (Parker and Koschmann 2018), all of which shape their purchasing decisions. Consequently, retailers seek effective shelf layout strategies to convert shoppers into buyers, while manufacturers actively negotiate for prime shelf placements to enhance product visibility and expand market share. Among the various factors that define shelf

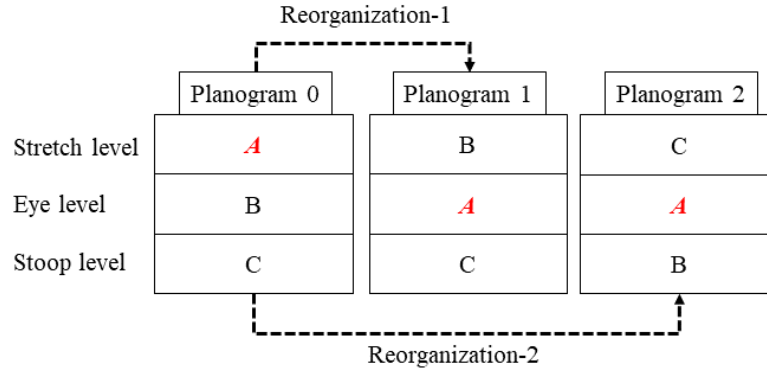


Figure 1 Illustration of moving Product A from stretch-level (in the planogram on the left) to eye-level.

layout (e.g., total space, vertical location, horizontal location, and number of facings), this study specifically focuses on the vertical shelf location of products.

Vertical location—the height at which a product is placed on a shelf (Chen et al. 2021)—directly affects product visibility in shelf layout. It is categorized into three generic zones: stretch-level, eye-level, and stoop-level. Eye-level shelves align with the average adult’s line of sight. Stretch-level shelves are placed above eye-level, requiring customers to reach up, whereas stoop-level shelves are below eye-level, requiring customers to bend down to access products (Ebster 2011). Grounded in the principles of consumer shopping behavior (Hoyer 1984, Hoch and Deighton 1989, Dickson and Sawyer 1990), Dreze et al. (1994) demonstrate that products sell best when placed at eye-level, a seminal finding that supports the well-known motto “eye-level is buy level” (Ausick 2017) and has served as the foundation for extensive research on planogram optimization. As a result, retailers strategically prioritize eye-level placement for high-margin products. To secure these prime locations, manufacturers often pay slotting fees (i.e., additional charges imposed by retailers), which account for over 16% of their new product introduction costs (Sudhir and Rao 2006).

Although using the eye-level location as a sales-boosting strategy is a well-established practice, the process by which retailers reorganize shelf layout to leverage the eye-level effect is not as straightforward. To illustrate this, consider a retailer selling three products on a shelving unit with the following initial planogram: Product A is placed on the stretch-level shelf, Product B on the eye-level shelf, and Product C on the stoop-level shelf. Now, suppose that the manufacturer of Product A would like to secure eye-level position to increase its sales. However, it remains unclear whether the resulting sales increase depends on the way Product A is placed at eye-level. As shown in Figure 1, the retailer can place Product A at eye-level in two ways: either (i) simply swap Products A and B, placing Product A at the eye-level and moving Product B to the stretch-level

shelf (i.e., Reorganization-1) or (ii) shift all three products, moving Product C to stretch-level, Product B to stoop-level, and Product A to eye-level (i.e., Reorganization-2). Both approaches result in Product A being placed at eye-level, yet they create two distinct planograms. The key question is then: How does the sales impact of moving Product A to eye-level differ between the two reorganizations? Surprisingly, there is no empirical study addressing this question.

Understanding whether different reorganization strategies yield distinct sales outcomes is critical for both retailers and manufacturers. For retailers, eye-level space is a scarce and highly monetized resource: misjudging the sales impact of one reorganization versus another could lead to suboptimal category performance, reduced margins, or strained manufacturer relationships. For manufacturers, slotting fees and trade allowances represent a substantial investment—exceeding \$50 billion annually (Rivlin 2016)—which are justified only if the promised eye-level placement delivers the expected return. If the realized sales lift depends on how products are rearranged, then manufacturers may overpay or underpay for eye-level guarantees, while retailers risk underdelivering on contractual expectations. Consequently, clarifying the differential effects of reorganization strategies can help both parties make more informed negotiations, align incentives, and ultimately design shelf layouts that maximize joint profitability. Motivated by this, our central research question is: How does planogram reorganization moderate the eye-level effect?

Our research relates to the literature on vertical location in shelf layout, which provides ample empirical evidence for the eye-level effect. Most studies rely on controlled laboratory experiments, with a few exceptions using secondary data. Experimental studies typically use eye-tracking technology in stores to examine visual attention toward products on different shelves or virtual retail simulations to assess consumer responses to vertical product placement. These studies show that visual attention increases as products move closer to eye-level. This increased attention has been shown to influence sales (Xu et al. 2023), brand evaluations (Chandon et al. 2009), and price and quality perceptions (Valenzuela and Raghurir 2015), though it diminishes as customers move through the store (Chen et al. 2021) or when store layouts lack cues about product popularity (Valenzuela et al. 2013). Empirical studies using secondary data from grocery retailers also confirm the eye-level effect (Dreze et al. 1994, Chung et al. 2007, Van Nierop et al. 2008, Xu et al. 2023). However, the existing literature does not examine whether different planogram reorganizations—despite placing the same product at eye-level—alter the strength of the eye-level effect. Even when we turn to analytical research focused on optimizing retail shelf allocation (e.g., Hansen et al. 2010, Russell and Urban 2010, Bianchi-Aguir et al. 2016, Smirnov and Huchzermeier 2019, Hübner et al.

2021, Gencosman and Begen 2022), this gap remains unaddressed. Our study contributes to this body of research by investigating how planogram reorganization moderates the eye-level effect.

To address our research question, we use data from a subset of a field experiment conducted in 2022 at one of the largest North American convenience-store retail chains. The original experiment evaluated alternative planograms across 818 products displayed on stretch-level, eye-level, and stoop-level shelves within 14 shelving units (i.e., fridges) per store, across three treatment and three control stores. Only two of the three treatment stores implemented planogram changes relevant to our research question. In these stores, products originally positioned at stretch- or stoop-levels were placed at eye-level using two distinct planogram reorganizations (Reorganization-1 and Reorganization-2, as illustrated in Figure 1). Therefore, our study focuses on planogram changes in these two treatment stores. In the control stores, the vertical locations of the same products in the same shelving units remained unchanged. Accordingly, we include all three control stores in our study. Because the interventions in the portion of the experiment we analyze constitute exogenous shocks to customers, we are able to employ an identification strategy to estimate the causal effect of two distinct planogram reorganizations on the eye-level effect.

Our research makes several fundamental contributions. First, our findings reveal that the eye-level effect depends on planogram reorganization strategy:

- Moving products from stretch-level to eye-level increased sales by 5.6% in Reorganization-1 but did not significantly impact sales in Reorganization-2.
- Moving products from stoop-level to eye-level increased sales by 10% in Reorganization-1, yet again, did not significantly change sales when executed under Reorganization-2.

We note that these findings emerge even though the well-established eye-level effect in the literature (Dreze et al. 1994, Van Nierop et al. 2008) is also evident in our empirical setting. When we consider sales across all stores throughout the study, products placed at eye-level shelves sold, on average, 6.9% more than those at stoop-level and 4.4% more than those at stretch-level. Collectively, these results suggest that the eye-level effect reported in the literature represents the average impact of eye-level placement, while our research uniquely uncovers how different planogram reorganizations moderate this effect, highlighting the theoretical significance of our study.

Second, we explain the underlying mechanism driving these results by what we call *partial visual search*. The sequential sampling models in psychology, coupled with eye-tracking experiments on visual search process, suggest that in the vertical dimension, customers (i) typically search for products sequentially, starting at eye-level, followed by stretch-level and stoop-level, and (ii) may end their search at any stage in this sequence, at which point they make a purchase decision

(Chandon et al. 2009, Lu and Hutchinson 2025). This implies that in-store grocery shoppers engage in different levels of visual search. Some customers may focus only on eye-level products (i.e., minimal visual search), others may consider eye-level and stretch-level products (i.e., partial visual search), and some may evaluate products across all shelves (i.e., full visual search). Comparing the two planograms obtained with different reorganizations, the choice sets remain the same for minimal visual searchers (i.e., who only see Product A in Planograms 1 and 2 in Figure 1) and full visual searchers (i.e., who consider Products A, B, and C in Planograms 1 and 2 in Figure 1). However, partial visual searchers experience a change in their choice set (i.e., they see Products A and B in Planogram 1, yet Products A and C in Planogram 2 in Figure 1). If, for these customers, the preference between the stretch-level and eye-level products differs across the two planograms, then the impact of eye-level placement will also vary between the two reorganizations. We provide empirical support for this mechanism through a mediation analysis. Using pre-intervention data, we quantify the change in preference within the choice set of partial visual searchers (i.e., eye-level and stretch-level products) using the Herfindahl-Hirschman Index (HHI). A decrease in HHI in our empirical data implies that products moved to eye-level are preferred, and thus will have more sales share. We find that Reorganization-1 decreased HHI, which led to a sales increase for products moved to eye-level, whereas Reorganization-2 had no impact on HHI, leaving sales unaffected.

Third, when examining all products across the three shelves, changes in the eye-level effect due to different planogram reorganizations directly translate into changes in overall sales. A shelving unit-level analysis reveals that Reorganization-1 increased overall sales across the treated shelves by 3.6% when moving products from stretch-level to eye-level and by 3.8% when moving products from stoop-level to eye-level. In contrast, Reorganization-2 had no impact on overall sales.

Lastly, while our aggregate findings imply that Reorganization-1 was, on average, more beneficial for the retailer, the underlying mechanism suggests that some shelving units could benefit more from Reorganization-2, specifically, if it results in a greater decrease in HHI than Reorganization-1. To explore this, we estimate shelving unit-level treatment effects. Our results show that Reorganization-2 generated a stronger eye-level effect than Reorganization-1 for 21% of shelving units, where products moved from stretch-level to eye-level, and for 14% of shelving units, where products moved from stoop-level to eye-level. This suggests that tailoring shelving reorganizations based on our identified mechanism could create opportunities for retailers to boost sales. To this end, we conduct a counterfactual analysis comparing three approaches: (i) assuming no difference between the two reorganizations, (ii) applying a blanket strategy based on aggregate findings (i.e., using Reorganization-1 for all shelving units), and (iii) tailoring reorganization for

each shelving unit as suggested by our shelving unit-level analysis. We find that, compared to the assumption of no difference between the two reorganizations, sales increase by 3.85% when shelving units are reorganized as in Reorganization-1 and by 4.32% when reorganization is tailored to each shelving unit. Overall, these results suggest that when moving products to eye-level locations, retailers should carefully assess potential preferences between the stretch- and eye-level products within each shelving unit and choose the reorganization type accordingly.

2. Empirical Setting

Our empirical setting represents a subset of a field experiment conducted in 2022 at a North American convenience store (C-store) retail chain that has an extensive presence in multiple countries, encompassing thousands of stores. The original experiment (described in Appendix A) was conducted in six randomly selected C-stores—three treatment and three control—in a metropolitan area, where our partner company has over 100 stores. These stores operate 24 hours a day, seven days a week. Strategic store-level decisions (i.e., product assortments, pricing, planogram design, and promotions) are centrally made by the regional marketing and operations teams, whereas day-to-day decisions (i.e., inventory and ordering) are made in a decentralized manner by store managers. The planograms follow a straightforward, expert-driven approach rather than data-driven optimization; for example, each shelving unit in a store is assigned to a specific product category.

The retailer offers 1,500 stock-keeping units (SKUs) across 35 product categories. The original experiment focused on products displayed in 14 fridges per store, including juices, water, energy drinks, sports drinks, carbonated soft drinks, beer, and wine. These products account for 38% of all offerings and 45% of the overall sales, allowing us to examine the eye-level effect across a broad range of products. Fridges have six shelves and the experiment examined the rearrangement of products on the first (from the top), second, and fifth shelves. The heights from the floor are 72" for the first shelf, 60" for the second shelf, and 24" for the fifth shelf. Based on consumer height distribution, we classify the first shelf as *stretch-level*, the second shelf as *eye-level*, and the fifth shelf as *stoop-level* throughout the paper. These classifications align with the commonly known categories of stretch-level, eye-level, and stoop-level in the literature (Chen et al. 2021).

As explained in Appendix A, only two of the three treatment stores implemented planogram changes that enable a comparison between Reorganization-1 and Reorganization-2. Thus, as illustrated in Figure 2, our analysis focuses on these two treatment stores and all three control stores from the original experiment. In the treatment stores, we observe eye-level placement across 14

		Pre-intervention	Post-intervention		Follow-up	
		Round-0 Mar 21 – Apr 16	Round-1 Apr 18 – May 28	Round-2 May 30 – Jun 18	Round-3 Jun 20 – Jul 09	Round-4 Jul 11 – Aug 06
Treatment Store 1	Stretch-level	P0 A	P1 B	P2 C	P4 B	P0 A
	Eye-level	B	A	A	C	B
	Stoop-level	C	C	B	A	C
Treatment Store 2	Stretch-level	P0 A	P3 A	P4 B	P0 A	P2 C
	Eye-level	B	C	C	B	A
	Stoop-level	C	B	A	C	B
Control Stores	Stretch-level	P0 A	P0 A	P0 A	P0 A	P0 A
	Eye-level	B	B	B	B	B
	Stoop-level	C	C	C	C	C

Figure 2 Empirical setting (shelves highlighted in bold, italic, and red indicate the treated shelves).

fridges, achieved through two different reorganizations: in Treatment Store 1 (originally labeled Treatment Store B), products originally placed at stretch-level were moved to eye-level, while in Treatment Store 2 (originally labeled Treatment Store C), products initially located at stoop-level were repositioned to eye-level (highlighted in bold, italic, and red in Figure 2). The products in each fridge, their original horizontal locations, and their number of facings remained constant across all five stores throughout the study, with only their vertical positions varying in treatment stores. This fully controlled setup enables us to isolate the effect of eye-level placement.

Round-0 in Figure 2 represents the baseline period with the original planogram P0 across the same 14 fridges in all stores. During Round-1, in Treatment Store 1, products were moved from stretch- to eye-level creating Planogram 1 (P1), while in Treatment Store 2, products were moved from stoop- to eye-level, creating Planogram 3 (P3). In Round-2, products at eye-level were maintained while products at stretch- and stoop-levels were swapped within the treatment stores, resulting in Planogram 2 (P2) in Treatment Store 1 and Planogram 4 (P4) in Treatment Store 2.

Hence, for products initially at stretch-level in Treatment Store 1 during Round-0 (i.e., Products A), Round-1 corresponds to eye-level placement with Reorganization-1, and Round-2 corresponds to eye-level placement with Reorganization-2. Similarly, for products initially at stoop-level in Treatment Store 2 during Round-0 (i.e., Products C), Round-1 corresponds to eye-level placement with Reorganization-1 and Round-2 corresponds to eye-level placement with Reorganization-2. From this perspective, this empirical setting is akin to a mixed design so that the products targeted for eye-level placement vary between stores (i.e., stretch- or stoop-level

products), while the type of planogram reorganization (i.e., Reorganization-1 or Reorganization-2) varies within stores.

The experiment continued with Rounds 3 and 4, testing alternative planograms within the treatment stores. While those rounds are not directly part of our main analysis, we use their data to assess the robustness of our main findings based on the first three rounds, as detailed in Section 4.3.

3. Data

We obtained transactional data, including the number of items sold per transaction, product price, and discounts, under a signed nondisclosure agreement. These data were provided exclusively for products displayed on stretch-level, eye-level, and stoop-level shelves during the study.

In the original experiment, planograms were changed by moving a set of products from one shelf to another while keeping their original horizontal locations and number of facings constant. Accordingly, we define the unit of analysis as a triplet of product set (the set of products placed on a specific shelf), store, and day (denoted by i , j , and t , respectively) and aggregate the transactions over triplets to obtain sales. Our dependent variable, $Sales_{ijt}$, is the natural logarithm of the average number of items sold per facing in product set i at store j on day t . This structure allows us to estimate the average eye-level effect across all number of facings and horizontal locations.

We use several control variables in our analyses. While products' prices were consistent across stores, it may vary over time. Therefore, at the product-level, we control for the price of all the products in product set i on day t ($Price_{it}$) by using the average price of all unique SKUs included in that set. We also control for promotional activities in a fridge using two variables: $SetPromotion_{it}$ represents the average monetary value of discounts across all unique SKUs present in product set i on day t , whereas $FridgePromotion_{it}$ represents the average monetary discount value across all unique SKUs displayed on all other shelves in the same fridge on day t (excluding products in product set i). Using pictures taken at each store on a daily basis during the original experiment, we are able to identify whether the lack of sales observation for a product on a specific day is due to stockout or the absence of demand. Using this identifier, we control for stockouts in a fridge using two variables: $SetStockout_{it}$ represents the number of SKUs that are out of stock in product set i on day t , while $FridgeStockout_{it}$ represents the number of SKUs that are out of stock on all other shelves in the same fridge on day t (excluding products in product set i). The variable $NumFaces_{it}$ represents the number of faces in product set i on day t . We also control for differences in product categories using indicator variables representing the category where the products in product set i belong to ($ProdCat_i$). At the store-level, we control for potential unobserved differences across

stores with fixed effects using an indicator variable representing store j that product set i belongs to ($Store_j$). Finally, we control for seasonality using four fixed effects: $Holiday_t$ represents whether day t is a national holiday, $Weekend_t$ represents whether day t is a weekend day, DoW_t represents the day of the week, and $Month_t$ represents the month indicators.

To ensure data integrity, we exclude all daily observations from a fridge if more than 30% of its products (including those placed on shelves not considered in our study) are out of stock. This exclusion accounts for approximately 10% of all observations. A t -test confirms that the removed observations are statistically balanced between the treatment and control groups. Our findings remain robust even when including these excluded observations.

For the main analysis, we focus on the shelves moved to eye-level in Treatment Store 1 (i.e., product set A) and Treatment Store 2 (i.e., product set C), along with all three shelves from the control stores in Rounds 0 to 2, representing a longitudinal panel data containing 10,322 daily observations across five stores. Table 1 presents descriptive statistics for this sample. With a mean variance inflation factor score of 1.47 and a range between 1.00 and 1.91, both below the commonly accepted threshold of 10 (Neter et al. 1996), there is little evidence of multicollinearity in our dataset. The remaining data are used in several robustness checks as described in Section 4.3.

4. Empirical Analysis

As detailed in Appendix A, the original experiment featured several design elements that strengthen causal inference: (i) stores were randomly selected, (ii) treatment status was randomly assigned at the store level, (iii) the planogram interventions were exogenous to customers, and (iv) implementation compliance was ensured through daily in-store monitoring. Moreover, because the intervention involved rearranging all products originally located at stretch-, eye-, and stoop-level shelves across all 14 fridges in each store, there was no discretionary selection of fridges or products. As a result, store selection, treatment assignment, shelving-unit selection, product selection, and the interventions themselves were all exogenous and orthogonal to realized sales outcomes. Together, these features alleviate the endogeneity concerns that typically arise in observational studies and provide a credible basis for causal interpretation, placing our setting closer to a randomized field experiment than to a standard observational study. In Section 4.1, we describe the identification strategy that leverages the exogenous variation generated by the original experiment. Section 4.2 presents the estimation results, and Section 4.3 examines the robustness of our findings.

Table 1 Descriptive statistics.

	Round-0						Round-1					
	Control		Treatment 1		Treatment 2		Control		Treatment 1		Treatment 2	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1. Sales	0.53	0.38	0.33	0.34	0.17	0.26	0.57	0.39	0.41	0.35	0.32	0.43
2. Price	1.68	0.49	1.51	0.29	2.00	0.57	1.75	0.46	1.63	0.26	2.06	0.53
3. SetPromotion	0.44	0.44	0.49	0.40	0.25	0.41	0.77	0.52	0.69	0.45	0.58	0.57
4. FridgePromotion	0.42	0.36	0.31	0.30	0.36	0.33	0.72	0.40	0.62	0.43	0.71	0.43
5. SetStockout	0.44	0.52	0.38	0.45	0.16	0.35	0.58	0.49	0.52	0.49	0.37	0.44
6. FridgeStockout	0.58	0.34	0.47	0.24	0.23	0.21	0.69	0.30	0.67	0.29	0.43	0.33
7. NumFaces	5.88	0.34	6.57	1.26	4.81	1.40	5.70	1.48	6.12	1.29	4.64	1.60
8. Holiday	0.03	0.18	0.03	0.17	0.04	0.19	0.05	0.22	0.05	0.22	0.05	0.21
9. Weekend	0.26	0.44	0.26	0.44	0.26	0.44	0.28	0.45	0.28	0.45	0.28	0.45
Sample Size	3,142						4,879					

	Round-2					
	Control		Treatment 1		Treatment 2	
	Mean	SD	Mean	SD	Mean	SD
1. Sales	0.62	0.40	0.42	0.36	0.33	0.37
2. Price	1.77	0.46	1.61	0.26	2.03	0.54
3. SetPromotion	0.78	0.50	0.70	0.42	0.49	0.55
4. FridgePromotion	0.73	0.42	0.65	0.44	0.73	0.45
5. SetStockout	0.40	0.50	0.31	0.42	0.35	0.45
6. FridgeStockout	0.48	0.42	0.53	0.26	0.59	0.41
7. NumFaces	6.02	1.70	6.50	1.38	4.52	1.68
8. Holiday	0.13	0.34	0.14	0.35	0.11	0.31
9. Weekend	0.27	0.44	0.29	0.45	0.28	0.45
Sample Size	2,301					

4.1. Identification Strategy

We identify the effect of eye-level placement with two different reorganizations as the post-intervention changes in sales at treatment group stores relative to that at control group stores. More formally, to address our main research question, we deploy a difference-in-differences (DiD) model with two treatment groups and two interventions. By estimating two differences between treatment and control groups, the DiD model addresses two key concerns common to most empirical methods. First, by estimating the difference in sales across time within each group, it mitigates the omitted variable bias caused by time-invariant unobserved heterogeneity (Card and Krueger 1994). Second, by estimating the difference in group-specific sales changes obtained in the first step, it accounts for and eliminates trend-specific differences between the treatment and control groups, ensuring a more robust identification of the treatment effect. Utilizing our longitudinal panel data

structure, we specify the DiD model (and the models in all subsequent analyses) as a fixed-effect model because the Hausman test suggests that the random effect formulation is not consistent:

$$\begin{aligned}
Sales_{ijt} = & \beta_0 + \beta_1 Treatment1_j \times Reorg1_t + \beta_2 Treatment1_j \times Reorg2_t \\
& + \beta_3 Treatment2_j \times Reorg1_t + \beta_4 Treatment2_j \times Reorg2_t \\
& + \beta_5 Reorg1_t + \beta_6 Reorg2_t + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt},
\end{aligned} \tag{1}$$

where $Treatment1_j$ represents whether store j is Treatment Store 1 (i.e., $Treatment1_j = 1$) or not (i.e., $Treatment1_j = 0$) and $Treatment2_j$ represents whether store j is Treatment Store 2 (i.e., $Treatment2_j = 1$) or not (i.e., $Treatment2_j = 0$). Thus, stores where both treatment variables are zero (i.e., $Treatment1_j = 0$ and $Treatment2_j = 0$) represent the control stores and serve as the reference condition. $Reorg1_t$ represents whether an observation belongs to Round-1 ($Reorg1_t = 1$) or not (i.e., $Reorg1_t = 0$) and $Reorg2_t$ represents whether an observation belongs to Round-2 ($Reorg2_t = 1$) or not (i.e., $Reorg2_t = 0$). $TVControls_{ijt}$ represents all time-variant control variables, including *Price*, *SetPromotion*, *FridgePromotion*, *SetStockout*, *FridgeStockout*, and *NumFaces*, $Seasonality_t$ represents the four seasonality fixed effects, u_{ij} represents the fixed effect for product set i in store j , and ϵ_{ijt} is the random error term. We account for heteroskedasticity in Equation (1) (and in all other estimations across all analyses) by using robust standard errors clustered at the product set-level. The coefficients of interest (i.e., the DiD estimators) are β_1 , β_2 , β_3 , and β_4 . These coefficients represent the causal effects of moving a product set (i) from stretch-level to eye-level with Reorganization-1 (i.e., β_1), (ii) from stretch-level to eye-level with Reorganization-2 (i.e., β_2), (iii) from stoop-level to eye-level with Reorganization-1 (i.e., β_3), and (iv) from stoop-level to eye-level with Reorganization-2 (i.e., β_4).

The fundamental assumption of the DiD model is the parallel trends assumption, which implies that, in the absence of intervention, the trend in the outcome variable should be the same between the treatment and control groups. To partially examine this assumption, we estimate the following model using data from the pre-intervention period (Round-0), during which both treatment and control stores operated under the same planogram (P0):

$$\begin{aligned}
Sales_{ijt} = & \theta_0 + \theta_1 Treatment1_j \times Trend_t + \theta_2 Treatment2_j \times Trend_t + \theta_3 Trend_t \\
& + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt},
\end{aligned} \tag{2}$$

where $Trend_t$ counts the days from the first day of Round-0 to day t . A statistically significant θ_1 (or θ_2) would indicate that the pre-intervention sales trend is different between Treatment Store 1 (or

Treatment Store 2) and the control stores, hence violating the parallel trends assumption. Table 2 demonstrates the results from the estimation of Equation (2). Model-1 estimates the coefficients of interest without the control variables. Model-2 includes all time-variant control variables, and Model-3 adds seasonality fixed effects. Across all models, we find that both θ_1 and θ_2 are statistically insignificant. This implies that the sales trends are similar across the two groups when operating under the same planogram (P0), providing partial support for the parallel trends assumption.

Table 2 Estimation results for pre-intervention trends.

	Model-1	Model-2	Model-3
<i>Treatment1</i> $\times$ <i>Trend</i>	-0.001 (0.003)	-0.001 (0.003)	-0.002 (0.003)
<i>Treatment2</i> $\times$ <i>Trend</i>	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
<i>Trend</i>	0.007*** (0.001)	0.008*** (0.001)	0.006*** (0.002)
<i>TVControls</i>	No	Yes	Yes
<i>Seasonality</i>	No	No	Yes
Sample size (N)	3,142	3,142	3,142
R^2	0.06	0.09	0.12

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2. Results

Table 3 reports estimates from Equation (1). Model-1 excludes control variables, Model-2 adds time-variant control variables, and Model-3 further includes seasonality fixed effects. Log-likelihood ratio tests for nested models (LR) confirm that each sequential refinement, from Model-1 to Model-3, significantly improves model fit. Accordingly, we use Model-3 to assess the treatment effects.

We find that the DiD estimator for Treatment Store 1 is positive and statistically significant for the Reorganization-1 ($\beta_1 = 0.056$), but insignificant for Reorganization-2 ($\beta_2 = -0.005$). This indicates that moving a set of products from stretch-level to eye-level increased sales for those products by 5.6% when implemented using Reorganization-1, yet did not significantly change their sales when implemented using Reorganization-2. Similarly, the DiD estimator for Treatment Store 2 is positive and statistically significant for Reorganization-1 ($\beta_3 = 0.100$), but insignificant for Reorganization-2 ($\beta_2 = -0.020$). This suggests that moving a set of products from stoop-level

Table 3 Eye-level effect estimation results.

	Model-1	Model-2	Model-3
<i>Treatment1</i> $\times$ <i>Reorg1</i>	0.048* (0.020)	0.066*** (0.020)	0.056** (0.019)
<i>Treatment1</i> $\times$ <i>Reorg2</i>	-0.012 (0.023)	0.005 (0.023)	-0.005 (0.023)
<i>Treatment2</i> $\times$ <i>Reorg1</i>	0.114*** (0.020)	0.105*** (0.020)	0.100*** (0.020)
<i>Treatment2</i> $\times$ <i>Reorg2</i>	0.055* (0.025)	0.028 (0.025)	0.020 (0.024)
<i>Reorg1</i>	0.043*** (0.008)	0.025* (0.012)	-0.064*** (0.014)
<i>Reorg2</i>	0.106*** (0.009)	0.114*** (0.013)	-0.007 (0.024)
<i>TVControls</i>	No	Yes	Yes
<i>Seasonality</i>	No	No	Yes
Sample size (N)	10,322	10,322	10,322
R^2	0.16	0.16	0.17
LR	-	322.64***	674.73***

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

to eye-level increased sales for those products by 10% when implemented using Reorganization-1, yet did not significantly change their sales when implemented using Reorganization-2.

To compare the effects of eye-level placement between the two reorganizations, we conduct a Wald test. We find that the two DiD estimators are significantly different for both Treatment Store 1 ($F(1, 10167) = 8.76$, $p - value = 0.003$) and Treatment Store 2 ($F(1, 10167) = 13.09$, $p - value < 0.001$). This provides evidence that the type of planogram reorganization (i.e., Reorganization-1 vs. Reorganization-2) significantly moderates the eye-level effect.

4.3. Robustness Tests

This subsection assesses the robustness of our findings against several alternative explanations.

4.3.1. Customer confusion. Frequent planogram rotations in the original experiment may have temporarily confused customers, causing them to experience difficulty locating their preferred products. Subsequently, they might have abandoned their purchase or opted for a substitute product. Once adapted to the new planograms, their original purchasing behavior could have resumed,

suggesting that observed differences in eye-level effect might reflect customer confusion rather than the planogram reorganization type itself.

To assess potential customer confusion, we consider Rounds 3 and 4, when the original planograms (P0) were reinstated, Round-4 at Treatment Store 1, and Round-3 at Treatment Store 2. If confusion affected sales, we would expect a significant difference in sales under P0 between the pre-intervention period (i.e., Round-0) and the follow-up period, despite the vertical product locations remaining unchanged. To quantify this effect, we used data from all three shelves in Treatment Store 1 (Rounds 0 and 4), in Treatment Store 2 (Rounds 0 and 3), and Control Stores (Rounds 0, 3, and 4)—totaling 10,659 observations—and estimated the following triple difference model:

$$\begin{aligned}
Sales_{ijt} = & \beta_0 + \beta_1 Treatment1_j \times Reorg4_t + \beta_2 Treatment2_j \times Reorg3_t \\
& + \beta_3 Eye - Level_{ij} \times Treatment1_j \times Reorg4_t + \beta_4 Eye - Level_{ij} \times Treatment2_j \times Reorg3_t \\
& + \beta_5 Stoop - Level_{ij} \times Treatment1_j \times Reorg4_t + \beta_6 Stoop - Level_{ij} \times Treatment2_j \times Reorg3_t \\
& + \beta_7 Reorg3_t + \beta_8 Reorg4_t + \beta_9 Eye - Level_{ij} \times Reorg3_t + \beta_{10} Eye - Level_{ij} \times Reorg4_t \\
& + \beta_{11} Stoop - Level_{ij} \times Reorg3_t + \beta_{12} Stoop - Level_{ij} \times Reorg4_t \\
& + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt},
\end{aligned} \tag{3}$$

where $Reorg3_t$ and $Reorg4_t$ indicate observations from Round-3 and Round-4, respectively, while $Eye - Level_{ij}$ and $Stoop - Level_{ij}$ denote whether product set i at store j was placed at eye-level or stoop-level under P0. In this specification, the coefficients β_1 and β_2 capture the difference in stretch-level product sales under P0 between the pre-intervention and follow-up periods at Treatment-Stores 1 and 2, respectively. The coefficients β_3 and β_4 indicate how these differences vary for eye-level products, while β_5 and β_6 reflect how they differ for stoop-level products.

The results, presented in Table 4, show that none of the coefficients of interest are statistically significant. Additionally, the estimated marginal effects for eye-level and stoop-level products (omitted for brevity) are also statistically insignificant. These findings indicate that sales at each shelf returned to pre-intervention levels when the original planograms were reinstated. This implies that there is no statistical evidence of customer confusion, confirming that our main findings are driven solely by the planogram reorganizations.

Table 4 Estimation results of triple difference model.

$Treatment1 \times Reorg4$	0.015 (0.025)
$Treatment2 \times Reorg3$	-0.009 (0.028)
$Eye - Level \times Treatment1 \times Reorg4$	-0.009 (0.036)
$Eye - Level \times Treatment2 \times Reorg3$	0.031 (0.039)
$Stoop - Level \times Treatment1 \times Reorg4$	0.010 (0.037)
$Stoop - Level \times Treatment2 \times Reorg3$	0.006 (0.040)
Sample size (N)	10,659
R^2	0.18

Notes: (i) Standard errors clustered at the shelving unit-level are reported in parentheses.

Notes: (ii) Only coefficients of interest are displayed for brevity.

(iii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3.2. Fixed intervention order. In the original experiment, the interventions produced four distinct post-intervention planograms across the treatment stores (P1 and P2 in Treatment Store 1; and P3 and P4 in Treatment Store 2). As a result, the two treatment stores did not share identical planograms within the same post-intervention round, thereby introducing a degree of randomization in the order of interventions within each store. However, one could argue that the procedure for generating the planograms was fixed: Round-1 always implemented Reorganization-1, and Round-2 always implemented Reorganization-2. In an ideal within-subject design—where the type of planogram reorganization varies within stores—the order of reorganizations would be randomized. The absence of such within-store randomization in the original experiment might raise the concern that the estimated treatment effects may be confounded with the order of reorganizations.

We assess potential order effect using data from Rounds 3 and 4. Specifically, we examine three cases where the same planogram appeared in both treatment stores, but in different rounds:

Setting-1: Planogram P0. P0 was reinstated in Round-4 at Treatment Store 1 and in Round-3 at Treatment Store 2. If order effects exist, sales change between Round-0 and the two follow-up rounds should differ significantly between the two stores. Recall that the analysis in section 4.3.1 (reported in Table 4) compares P0 in this setting. Wald tests comparing estimated marginal effects

reveal no significant differences between the two stores for stretch-level products ($F(1, 10405) = 0.50$, $p - value = 0.479$), eye-level products ($F(1, 10405) = 0.21$, $p - value = 0.644$), or stoop-level products ($F(1, 10405) = 0.59$, $p - value = 0.443$). Thus, no order effect is detected.

Setting-2: Planogram P4. P4 appeared in Round-3 at Treatment Store 1 and in Round-2 at Treatment Store 2. We focus on product set C in Treatment Store 2. If order effects exist, the change in sales for this set between P0 in Round-0 and P4 should differ across the treatment stores. To assess this, we estimate the following model and report the results under Model-1 in Table 5:

$$\begin{aligned} Sales_{ijt} = & \beta_0 + \beta_1 Treatment1_j \times Reorg3_t + \beta_2 Treatment2_j \times Reorg2_t \\ & + \beta_3 Reorg2_t + \beta_4 Reorg3_t + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt}. \end{aligned} \quad (4)$$

We find that moving products from stoop-level to eye-level did not significantly change their sales in either store, providing a sanity check for our results of eye-level placement effect of product set C under Reorganization-2 in the main analysis. More importantly, a Wald test shows no significant difference between the two stores in the eye-level placement effect for product set C ($F(1, 6813) = 0.01$, $p - value = 0.909$), implying that no order effect exists.

Setting-3: Planogram P2. P2 appeared in Round-2 at Treatment Store 1 and in Round-4 at Treatment Store 2. We focus on product set A in Treatment Store 1. If order effects exist, the sales change for product set A from P0 (pre-intervention) to P2 should differ across the two stores. To examine this, we estimate the following model and report the results under Model-2 in Table 5:

$$\begin{aligned} Sales_{ijt} = & \beta_0 + \beta_1 Treatment1_j \times Reorg2_t + \beta_2 Treatment2_j \times Reorg4_t \\ & + \beta_3 Reorg2_t + \beta_4 Reorg4_t + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt}. \end{aligned} \quad (5)$$

We find that moving products from stretch-level to eye-level did not significantly change their sales in either store, providing a sanity check for our results of eye-level placement effect of product set A under Reorganization-2 in the main analysis. A Wald test confirms no significant difference in the eye-level placement effect for product set A between the two stores ($F(1, 6813) = 0.12$, $p - value = 0.729$), again implying that no order effect exists. Taken together, evidence from these three settings indicates that the order of reorganizations does not confound treatment effects. Hence, our main results are unlikely to suffer from order effects.

4.3.3. Variation in duration of post-intervention rounds. In the original field experiment, post-intervention data collection spanned 42 days in Round-1 and 21 days in Round-2. One might argue that intervention effects only emerge after customers adapt to new planograms. Since

Table 5 Estimation results for intervention order effect.

	Model-1	Model-2
<i>Treatment1</i> $\times$ <i>Reorg3</i>	0.009 (0.028)	
<i>Treatment2</i> $\times$ <i>Reorg2</i>	0.013 (0.026)	
<i>Treatment1</i> $\times$ <i>Reorg2</i>		-0.007 (0.023)
<i>Treatment2</i> $\times$ <i>Reorg4</i>		0.004 (0.024)
<i>Reorg2</i>	0.124*** (0.021)	0.108*** (0.015)
<i>Reorg3</i>	0.177*** (0.018)	
<i>Reorg4</i>		0.175*** (0.017)
<i>TVControls</i>	Yes	Yes
<i>Seasonality</i>	Yes	Yes
Sample size (N)	6,979	8,417
R^2	0.17	0.17

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Round-2 lasted only half as long as Round-1, the difference in treatment effects between the two reorganizations could be attributed to insufficient observation time for Round-2. If Round-2 had been observed for 42 days instead of 21, similar treatment effects might have emerged.

To test this alternative explanation, we repeat our analysis excluding observations from the second 21 days in Round-1, ensuring an equal comparison with the 21-days duration of Round-2. If the alternative explanation holds true, the treatment effects for this subset of Round-1 should align with those from Round-2. Table 6 presents the results, showing that all four DiD estimators remain qualitatively consistent with our main findings. Additionally, the effects of eye-level placement between the two reorganizations remain significantly different in both Treatment Store 1 ($F(1, 7841) = 5.60$, $p - value = 0.0179$) and Treatment Store 2 ($F(1, 7841) = 13.25$, $p - value < 0.001$). These results confirm that the observed difference in treatment effects between the two reorganizations are not simply due to the shorter duration of Round-2, ruling out this alternative explanation.

Table 6 Estimation results for eye-level effect with equal rounds.

	Model-1	Model-2	Model-3
<i>Treatment1</i> $\times$ <i>Reorg1</i>	0.054* (0.023)	0.063** (0.023)	0.051* (0.023)
<i>Treatment1</i> $\times$ <i>Reorg2</i>	-0.011 (0.023)	0.008 (0.023)	-0.005 (0.023)
<i>Treatment2</i> $\times$ <i>Reorg1</i>	0.122*** (0.024)	0.118*** (0.024)	0.114*** (0.023)
<i>Treatment2</i> $\times$ <i>Reorg2</i>	0.054* (0.025)	0.037 (0.025)	0.021 (0.024)
<i>Reorg1</i>	0.012 (0.009)	-0.017 (0.013)	-0.089*** (0.015)
<i>Reorg2</i>	0.105*** (0.009)	0.093*** (0.014)	-0.027 (0.026)
<i>TVControls</i>	No	Yes	Yes
<i>Seasonality</i>	No	No	Yes
Sample size (N)	7,996	7,996	7,996
R^2	0.18	0.18	0.18
LR	-	247.75***	536.04***

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3.4. Do products sell best at eye-level? One may wonder whether our empirical setting differs from those in the existing literature regarding the well-established eye-level effect. If this effect were absent in our setting, our findings could be driven by this difference. To check this, we use data from all three shelves—including those in the treatment stores, excluded from our main analysis—across five stores (24,050 observations) and estimate the following regression model:

$$\begin{aligned}
Sales_{ijt} = & \beta_0 + \beta_1 Stretch - Level_{ijt} + \beta_2 Eye - Level_{ijt} + \beta_3 Reorg1_t + \beta_4 Reorg2_t \\
& + TVControls_{ijt} + Seasonality_t + u_{ij} + \epsilon_{ijt},
\end{aligned} \tag{6}$$

where $Stretch - Level_{ijt}$ denotes whether product set i at store j in time t is placed at stretch-level ($Stretch - Level_{ijt} = 1$) or not ($Stretch - Level_{ijt} = 0$) and $Eye - Level_{ijt}$ denotes whether product set i at store j in time t is placed at eye-level ($Eye - Level_{ijt} = 1$) or not ($Eye - Level_{ijt} = 0$). Accordingly, β_1 and β_2 represent the associations of sales with stretch-level placement and eye-level placement, respectively, relative to stoop-level placement.

Table 7 Estimation results for all vertical location effects.

	Model-1	Model-2
<i>Stretch – Level</i>	0.025* (0.012)	- -
<i>Eye – Level</i>	0.069* (0.014)	0.044*** (0.014)
<i>Stoop – Level</i>	- -	-0.025* (0.012)
<i>TVControls</i>	Yes	Yes
<i>Seasonality</i>	Yes	Yes
Sample size (N)	24,050	24,050
R^2	0.15	0.15

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model-1 in Table 7 demonstrates the results. We find that compared to products placed at stoop-level, those placed at eye-level sell 6.9% more on average, while products placed at stretch-level sell 2.5% more. To further assess whether eye-level placement significantly increases sales relative to stretch-level, we modify Equation (6) by replacing $Stretch - Level_{ijt}$ with $Stoop - Level_{ijt}$, which denotes whether product set i at store j in time t is placed at stoop-level ($Stoop - Level_{ijt} = 1$) or not ($Stoop - Level_{ijt} = 0$). As shown in Model-2 in Table 7, products placed at eye-level sell 4.4% more on average than those at stretch-level. Collectively, these findings confirm that the well-established eye-level effect reported in prior literature is also evident in our empirical setting, effectively ruling out this alternative explanation.

4.3.5. Alternative operationalization of dependent variable. We further assess the robustness of our results by examining an alternative operationalization of the dependent variable. In particular, we define sales as the natural logarithm of the average dollar sales (i.e., sales value) per facing for product set i at store j on day t and re-estimate our main model using this variable. As demonstrated in Table 8, while the magnitudes differ, the DiD estimators for both reorganizations at Treatment Store 1 and for Reorganization-1 at Treatment Store 2 remain qualitatively consistent with our main findings in Table 3. However, unlike our main results, we find that the DiD estimator for Reorganization-2 at Treatment Store 2 ($\beta_3 = 0.153$) is statistically significant. Despite these nuances, when comparing the effects of eye-level placement between the two reorganizations, we find that the two DiD estimators remain significantly different for both Treatment Store

Table 8 Estimation results for eye-level effect with sales value.

	Model-1	Model-2	Model-3
<i>Treatment1</i> $\times$ <i>Reorg1</i>	0.160*** (0.043)	0.187*** (0.044)	0.167*** (0.042)
<i>Treatment1</i> $\times$ <i>Reorg2</i>	-0.020 (0.050)	0.006 (0.051)	-0.015 (0.049)
<i>Treatment2</i> $\times$ <i>Reorg1</i>	0.319*** (0.044)	0.306*** (0.044)	0.296*** (0.043)
<i>Treatment2</i> $\times$ <i>Reorg2</i>	0.215*** (0.053)	0.171** (0.054)	0.153** (0.053)
<i>Reorg1</i>	0.070*** (0.016)	0.025 (0.025)	-0.147*** (0.030)
<i>Reorg2</i>	0.215*** (0.020)	0.219*** (0.028)	-0.026 (0.052)
<i>TVControls</i>	No	Yes	Yes
<i>Seasonality</i>	No	No	Yes
Sample size (N)	10,322	10,322	10,322
R^2	0.17	0.17	0.17
LR	-	168.24***	646.32***

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

1 ($F(1, 10167) = 16.54$, $p - value < 0.001$) and Treatment Store 2 ($F(1, 10167) = 8.69$, $p - value = 0.0032$). These results provide evidence that planogram reorganization type (i.e., Reorganization-1 vs. Reorganization-2) significantly moderates the eye-level effect, even when using dollar sales instead of sales quantity as our dependent variable. This confirms the robustness of our findings with respect to an alternative operationalization of sales.

In summary, our results remain robust across various alternative explanations and model specifications. Overall, our findings provide evidence that the eye-level effect is moderated by the broader planogram reorganization. Notably, this result emerges even though the well-established eye-level effect (i.e., products sell best when placed at eye-level) is also present in our data. In the next section, we explore the mechanism underlying our main findings.

5. Underlying Mechanism

We posit that the difference in the eye-level effect between the two reorganizations is driven by what we call consumers' partial visual search. In the following, we first explain our theory for this position, and then provide empirical evidence to support it.

5.1. Theory

Sequential sampling models from psychology explain how customers visually search for products and make purchase decisions (e.g., Weitzman 1979, Bronnenberg et al. 2016, Ursu et al. 2023, 2024, Lu and Hutchinson 2025, Ursu et al. 2025). These models suggest that customers accumulate information about products sequentially through a series of quick, discrete decisions. Initially, they gather information from a relatively small set of options and determine if a satisfactory choice exists within that set. If they find one, they end their search and make a purchase decision. Otherwise, they continue exploring additional products, expanding their choice set. This process continues sequentially over time until they either identify an ideal option or leave the search process if the cumulative cost for further search outweighs the expected utility of finding a better product.

Applying this model to shelving unit, we posit that customers are heterogeneous in forming choice sets during visual search. Some may continue acquiring information until they have assessed all products, considering every option available on the shelving unit in their choice set. Others may stop after evaluating only a few shelves, implying that their choice sets include only a subset of all products on the shelving unit. Therefore, even when two customers scan the same shelving unit, the choice sets they form, and ultimately base their purchase decision on, may differ.

The literature on vertical shelf location supports this proposition. Eye movement studies indicate that customers can quickly recognize the basic semantic characteristics of a shelving unit, such as product category and number of shelves, during their initial eye fixation, which lasts approximately 200-500 milliseconds (Henderson and Hollingworth 1999). However, detailed product information, which is necessary to evaluate a product as a choice, requires increased visual attention through additional fixations (Chandon et al. 2009). Research shows that customers give more visual attention to products at or near eye-level than those at other levels (Chen et al. 2021, Xu et al. 2023), suggesting that whether a customer includes a product in her choice set depends on its shelf placement. Research also shows that initial fixations tend to occur near the center of the shelving unit (i.e., at or near eye-level) and later shift to other adjacent shelves (Chandon et al. 2007). Collectively, these findings indicate that when searching for products across the three shelves focused in our study (i.e., the first, second, and fifth shelves), customers are likely to follow a sequential

search pattern, starting at eye-level (i.e., second shelf), moving to the adjacent stretch-level (first shelf), and finally looking at the stoop-level (fifth shelf). Integrating this search pattern with the sequential sampling model, three types of visual searchers are likely present in our study:

- Minimal visual searchers: These customers limit their visual search to products at eye-level, making purchase decisions solely from this selection.
- Partial visual searchers: These customers expand their choice set to include both eye-level and stretch-level products.
- Full visual searchers: These customers evaluate all products across the shelves before making purchase decisions.

Supporting this segmentation, in an eye-tracking experiment, Chandon et al. (2009) found that products at or near eye-level received the highest visual attention, followed by stretch-level products, whereas stoop-level products received the lowest attention.

When we compare the two planograms obtained after the two distinct reorganizations (i.e., planograms in Round-1 and Round-2 at treatment stores in Figure 2), we observe that

- the choice set remains unchanged across both rounds for minimal visual searchers, i.e., it includes product set A in Treatment Store 1 and product set C in Treatment Store 2,
- the choice set is different for partial visual searchers. In Treatment Store 1, it includes product sets A and B in Round-1, but product sets A and C in Round-2. In Treatment Store 2, it includes product sets A and C in Round-1, but product sets B and C in Round-2, and
- the choice set remains constant across both rounds for full visual searchers, i.e., it includes all three product sets (A, B, and C) in both treatment stores.

These observations suggest that post-intervention, while minimal and full visual searchers maintained the same choice sets across both rounds, partial visual searchers experienced a shift in their choice sets from Round-1 to Round-2 due to differences in stretch-level and eye-level product placements. If the proportion of partial visual searchers preferring stretch-level products over eye-level products changes between the two rounds, the effect of eye-level placement should also differ between the two reorganizations. This is the premise of our theoretical mechanism underlying our empirical results. In the next subsection, we provide empirical support for this mechanism.

5.2. Empirical Evidence

A shift in preference for eye-level products over stretch-level products changes the level of sales concentration among those products. Therefore, we operationalize the concentration of sales among stretch-level and eye-level products (i.e., the partial visual searcher's choice set) using the

Herfindahl-Hirschman Index (HHI), which is used to measure market concentration in an industry or assortment. As detailed in Appendix B, we calculate HHI for each of the 70 fridges across the three rounds (Rounds 0 to 2) using only pre-intervention data from Round-0. Relying on pre-intervention data ensures that the mediating variable is not derived from outcome variables observed after the intervention, thereby preventing potential reverse causality.

An increase in sales of products placed at eye-level can increase or decrease HHI, depending on how sales were distributed across stretch-level and eye-level products before the intervention. If sales increase more for products with lower sales shares (i.e., less than 50%) than those with higher sales shares (i.e., more than 50%), the choice set becomes more balanced, leading to a lower HHI. Conversely, if sales increase more for products with higher sales shares than those with lower sales shares, then sales in a choice set concentrate on certain products, leading to a higher HHI. In our pre-intervention data, most products had lower sales shares, suggesting that a negative correlation between sales and HHI is likely. We further justify this assumption by estimating a simple regression between pre-intervention sales and HHI. The result shows a strong negative correlation between sales and HHI (i.e., $\beta = -1.91$, $p\text{-value} < 0.001$, $R^2 = 0.24$).

Our theory suggests that the interventions (i.e., Reorganization-1 vs. Reorganization-2) can shift sales from stretch-level products to eye-level products at different degrees, which then causes the treatment effects. This implies that the extent of sales shift, and thus the change in HHI, mediates the effects of the interventions. To test this mechanism, we specify the following mediation model as a recursive system of equations:

$$\begin{aligned}
Sales_{ijkt} = & \beta_0 + \beta_1 Intervention11_{ijkt} + \beta_2 Intervention12_{ijkt} + \beta_3 Intervention21_{ijkt} \\
& + \beta_4 Intervention22_{ijkt} + \beta_5 HHI_{jkt} + \beta_6 Reorg1_t + \beta_7 Reorg2_t + TVControls_{ijkt} \\
& + Seasonality_t + u_{ijk} + \epsilon_{ijkt}, \\
HHI_{jkt} = & \theta_0 + \theta_1 Intervention11_{ijkt} + \theta_2 Intervention12_{ijkt} + \theta_3 Intervention21_{ijkt} \\
& + \theta_4 Intervention22_{ijkt} + \theta_5 Reorg1_t + \theta_6 Reorg2_t + v_{ijk} + \varepsilon_{ijkt},
\end{aligned} \tag{7}$$

$$\tag{8}$$

where $Intervention11_{ijkt}$ indicates whether an observation from product set i in fridge k at store j belongs to Round-1 in Treatment Store 1 ($Intervention11_{ijkt}=1$) or not ($Intervention11_{ijkt}=0$); $Intervention12_{ijkt}$ indicates whether an observation belongs to Round-2 in Treatment Store 1 ($Intervention12_{ijkt}=1$) or not ($Intervention12_{ijkt}=0$); $Intervention21_{ijkt}$ indicates whether an observation belongs to Round-1 in Treatment Store 2 ($Intervention21_{ijkt}=1$) or not

($Intervention21_{ijkt} = 0$); $Intervention22_{ijkt}$ indicates whether an observation belongs to Round-2 in Treatment Store 2 ($Intervention22_{ijkt}=1$) or not ($Intervention22_{ijkt} = 0$); HHI_{jkt} represents HHI for the stretch-level and eye-level products in fridge k at store j at time t (as calculated in Appendix B); u_{ijk} and v_{ijk} represent fixed effects; and ϵ_{ijkt} and ε_{ijkt} are the random error terms.

Equation (7) in this specification represents the DiD model with two-way fixed effects, yielding estimates identical to those obtained with the interaction terms in Equation (1). Thus, the DiD estimators in Equation (7) (i.e., β_1 , β_2 , β_3 , and β_4) represent the treatment effects on sales, while in Equation (8) (i.e., θ_1 , θ_2 , θ_3 , and θ_4), they represent the treatment effect on HHI. This specification allows us to quantify the extent to which the treatment effects are mediated by changes in HHI.

We present the results in Table 9. Model-1 reports the estimation results for Equation (7), which, as expected, are identical to those in Table 3. Model-2 shows the mediation model estimates. In the HHI model, we find that the DiD estimator for Treatment Store 1 is negative and significant for Reorganization-1 ($\theta_1 = -0.035$), but insignificant for Reorganization-2 ($\theta_2 = -0.001$). Similarly, for Treatment Store 2, the DiD estimator is negative and significant for Reorganization-1 ($\theta_3 = -0.062$), but insignificant for Reorganization-2 ($\theta_4 = -0.001$). These results suggest that moving products from stretch-level to eye-level in Treatment Store 1 and from stoop-level to eye-level in Treatment Store 2 decreased the HHI of the partial visual searcher's choice sets (i.e., made the choice sets more balanced) under Reorganization-1, but had no significant effect under Reorganization-2. Given that eye-level products are identical across both reorganizations, this finding is plausible only if sales shares shift from eye-level products to stretch-level products in Round-2 compared to Round-1, providing evidence of a decrease in preference for eye-level products between the two rounds.

In the Sales model of Model-2, the direct effects of treatments are insignificant ($\beta_1 = -0.013$, $\beta_2 = -0.012$, $\beta_3 = -0.014$, and $\beta_4 = 0.013$). However, the indirect effects of treatments on sales, mediated by HHI, are estimated as: $\alpha = 0.064$ ($p\text{-value} < 0.001$) for $Intervention11$, $\alpha = 0.002$ ($p\text{-value} > 0.1$) for $Intervention12$, $\alpha = 0.114$ ($p\text{-value} < 0.001$) for $Intervention21$, and $\alpha = 0.002$ ($p\text{-value} > 0.1$) for $Intervention22$. These findings confirm that the effects of the interventions on *Sales* are fully mediated by the change in HHI.

Overall, the mediation analysis reveals that compared to products placed at eye-level, products placed at stretch-level under Reorganization-2 are, on average, preferred more than those placed under Reorganization-1. This subsequently decreased the eye-level placement effect in Round-2 compared to Round-1, supporting the theoretical mechanism underlying our findings. In the next section, we explore the economic implications of these findings.

Table 9 Mediation analysis results.

	Model-1	Model-2
SALES MODEL	<i>Intervention11</i>	0.056** (0.019)
	<i>Intervention12</i>	-0.005 (0.023)
	<i>Intervention21</i>	0.100*** (0.020)
	<i>Intervention22</i>	0.020 (0.024)
	<i>HHI</i>	- (0.160)
	<i>TVControls</i>	Yes
	<i>Seasonality</i>	Yes
	<i>Intervention11</i>	- (0.001)
	<i>Intervention12</i>	- (0.001)
	<i>Intervention21</i>	-0.062*** (0.001)
HHI MODEL	<i>Intervention22</i>	-0.001 (0.001)
	Sample size (N)	10,322
		10,322

Notes: (i) Standard errors clustered at the product set-level are reported in parentheses.

(ii) The coefficients of *Reorg1* and *Reorg2* are not illustrated for brevity.

(iii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6. Economic Implications

To examine the economic implications of our findings, we first assess how the difference in the eye-level placement effect between the two reorganizations impacts overall sales across the three shelves focused in our study. Motivated by our mechanism testing, we posit that the comparison between these reorganizations may vary across fridges. While, on average, Reorganization-1 led to a greater eye-level effect than Reorganization-2, the opposite could hold at the shelving unit-level, particularly if Reorganization-2 results in a lower HHI for a fridge than Reorganization-1. In such cases, retailers would benefit from managing planogram reorganizations (i.e., Reorganization-1 vs.

Table 10 Estimation results for overall sales.

$Treatment1 \times Reorg1$	0.036*** (0.010)
$Treatment1 \times Reorg2$	-0.015 (0.012)
$Treatment2 \times Reorg1$	0.038*** (0.009)
$Treatment2 \times Reorg2$	-0.018 (0.012)
Sample size (N)	4,587
R^2	0.29

Notes: (i) Standard errors clustered at the shelving unit-level are reported in parentheses.
(ii) *Reorg1*, *Reorg2*, *TVControls*, and *Seasonality* are included in the estimation.
(iii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reorganization-2) at the shelving unit-level rather than applying a uniform strategy across all shelving units. To enhance our managerial insights, we next extend our analysis to compare the two reorganizations at the shelving unit-level and quantify the counterfactual impact on sales had the retailer tailored planogram reorganization to each fridge.

6.1. Effect of Reorganization-1 vs. Reorganization-2 on Overall Sales

To assess the impact on overall sales, we use data from all three shelves across the five stores, aggregating it at the shelving unit-level to obtain fridge-level sales. We operationalize sales as the natural logarithm of the average number of items sold per facing in fridge k at store j on day t and estimate Equation (1) using this outcome variable. Table 10 presents the results from this estimation. We find that sales across the three shelves increased significantly by 3.6% ($\beta = 0.036$) in Treatment Store 1 and by 3.8% ($\beta = 0.038$) in Treatment Store 2 under Reorganization-1. In contrast, Reorganization-2 had no significant impact on overall sales. These findings indicate that the type of reorganization affects overall sales in a manner consistent with its influence on the eye-level effect. This consistency suggests that changes in the eye-level effect resulting from different planogram reorganizations directly translate into changes in overall sales.

6.2. Shelving Unit-Level Analysis and Planogram Reorganization Implications

We examine the variation in the effect of planogram reorganization on eye-level placement across the shelving units by estimating the treatment effects for each fridge in isolation. To do so, we

estimate the following random-slope DiD model with two-way fixed effects, in which the DiD estimators vary between fridges:

$$\begin{aligned}
 Sales_{ijkt} = & \beta_0 + \beta_1 Intervention11_{ijkt} + \beta_2 Intervention12_{ijkt} + \beta_3 Intervention21_{ijkt} \\
 & + \beta_4 Intervention22_{ijkt} + \beta_5 HHI_{jkt} + \beta_6 Reorg1_t + \beta_7 Reorg2_t + TVControls_{ijkt} \\
 & + Seasonality_t + \xi_{0ijk} + \xi_{1ijk} Intervention11_{ijkt} + \xi_{2ijk} Intervention12_{ijkt} \\
 & + \xi_{3ijk} Intervention21_{ijkt} + \xi_{4ijk} Intervention22_{ijkt} + u_{ijk} + \epsilon_{ijkt},
 \end{aligned} \tag{9}$$

where (i) ξ_{0ijk} denotes the deviation of product set i 's (in fridge k at store j) intercept from the mean intercept of β_0 and (ii) ξ_{1ijk} , ξ_{2ijk} , ξ_{3ijk} , and ξ_{4ijk} denote the deviations of product set i 's slopes for the DiD estimators from the mean slopes of β_1 , β_2 , β_3 , and β_4 , respectively. Hence, in this specification, the eye-level effects in fridge k at store j for the two planogram reorganizations across the two treatment stores are estimated to be $\beta_1 + \xi_{1ijk}$, $\beta_2 + \xi_{2ijk}$, $\beta_3 + \xi_{3ijk}$, and $\beta_4 + \xi_{4ijk}$.

Table 11 presents the results from the estimated model. The significant χ^2 test statistic confirms that eye-level effects vary significantly across fridges. The estimated coefficients β_1 , β_2 , β_3 , and β_4 align with our main findings in Table 3, providing a sanity check for this specification. Fridge-specific estimates indicate that Reorganization-2 resulted in a greater eye-level effect than Reorganization-1 for 3 out of 14 fridges (i.e., 21% of shelving units) in Treatment Store 1 and for 2 out of 14 fridges (i.e., 14% of shelving units) in Treatment Store 2.

These findings suggest that when moving a product to eye-level in a shelving unit, customizing planogram reorganizations to that shelving unit, rather than applying the same reorganization to all shelving units, could enable retailers to boost overall sales. To quantify the impact of this approach, we conduct a counterfactual analysis, as presented in Appendix C. Our counterfactual analysis considers three scenarios. In the first scenario (i.e., random reorganization), the retailer assumes no difference between the two reorganizations, randomly selecting one for each shelving unit. In the second scenario (i.e., uniform reorganization), the retailer follows the aggregate findings and applies Reorganization-1 to all shelving units. In the third scenario (i.e., tailored reorganization), the retailer customizes reorganization for each shelving unit based on our shelving unit-level analysis. We find that, compared to the random reorganization, the uniform reorganization could have increased overall sales by 3.85%, while the tailored reorganization could have led to a 4.32% increase. These results underscore the economic significance of our findings.

Table 11 Shelving unit-level analysis results.

<i>Intervention11</i>	0.059*
	(0.024)
<i>Intervention12</i>	-0.002
	(0.027)
<i>Intervention21</i>	0.103***
	(0.028)
<i>Intervention22</i>	0.027
	(0.037)
<i>TVControls</i>	Yes
<i>Seasonality</i>	Yes
$\sigma_{\xi_{1ijk}}$	0.010***
	(0.026)
$\sigma_{\xi_{2ijk}}$	0.059*
	(0.028)
$\sigma_{\xi_{3ijk}}$	0.073**
	(0.025)
$\sigma_{\xi_{4ijk}}$	0.105**
	(0.033)
Sample size (N)	10,322
$\chi^2(5)$	2,395***

(i) The coefficients of *Reorg1* and *Reorg2* are not illustrated for brevity.

(ii) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7. Conclusion

The notion that “eye-level is buy level” serves as the foundation for planogram optimization in retail and influences contractual agreements between retailers and manufacturers. However, no study has empirically tested whether shifting a product to an eye-level shelf increases its sales by a consistent amount, regardless of how the overall planogram is reorganized to accommodate that move. To investigate this, we study an empirical setting that represents a subset of a field experiment conducted in collaboration with a North American convenience store chain across five convenience stores. Our empirical setting focuses on two interventions at two treatment stores, simulating eye-level placement for the same product set by two different planogram reorganizations.

From a theoretical perspective, our study demonstrates that planogram reorganization moderates the eye-level effect. In particular, we find that moving products from stretch-level to eye-level

increased sales by 5.6%, while moving them from stoop-level to eye-level boosted sales by 10% under Reorganization-1. However, the same eye-level placements did not result in statistically significant sales changes under Reorganization-2. Notably, these findings hold even when the eye-level effects reported in the literature (e.g. Dreze et al. 1994, Van Nierop et al. 2008) were also observed across the five stores in our data. That is, products placed at eye-level, on average, sold significantly more than products placed at stretch-level and stoop-level. This suggests that while the eye-level effect reported in prior literature represents the average impact of eye-level placement, our results indicate that the type of planogram reorganization does moderate this effect.

We attribute these findings to partial visual search. Customers likely exhibit heterogeneity in how they search for products on a shelving unit. For those who search all products or focus solely on eye-level products, the two types of planogram reorganization (Reorganization-1 vs. Reorganization-2) result in identical choice sets. However, for customers who primarily consider eye-level and stretch-level products (i.e., partial visual searchers), these reorganizations create two distinct choice sets. If these customers prefer eye-level products over stretch-level products at varying degrees under the two planograms, then the impact of eye-level placement varies accordingly, even when the same products are placed at eye-level in both planograms. Focusing on the change in preference between eye-level and stretch-level products, our empirical analysis provides support for this mechanism.

Beyond the theoretical implications, our findings also have important economic consequences. Specifically, we find that differences in the eye-level effect between the two planogram reorganizations may directly influence overall sales. This suggests that the planogram reorganization type to operationalize the eye-level placement can affect not only the magnitude of eye-level effect, but also overall sales of the entire planogram. However, our shelving unit-level analysis reveals that customizing the type of planogram reorganization for each shelving unit, by accounting for shelving unit-level preference patterns between eye-level and stretch-level products, can further enhance sales. A counterfactual analysis based on our data indicates that when moving products from stretch-level or from stoop-level to eye-level, compared to assuming no difference between the two reorganizations, sales across the three treated shelves increase by 3.85% under Reorganization-1 and by 4.32% when the reorganization is tailored to each shelving unit.

From a practical standpoint, our study offers several managerial insights. First, our analysis of the underlying mechanism indicates that the eye-level effect is amplified when adjacent shelves contain less-preferred products. Accordingly, eye-level placement is more likely to generate the expected return under contractual arrangements that grant a manufacturer exclusive or near-exclusive control of a shelving unit (e.g., category captaincy or exclusive shelf-space agreements), since such

exclusivity effectively limits competitors' ability to occupy adjacent locations with more-preferred products (FTC 2001). Second, in contracts where retailers retain full discretion over planogram design, our findings suggest that managers should carefully evaluate relative product preferences across stretch- and eye-level locations before choosing the reorganization strategy. This ensures that moving a product to eye-level not only meets contractual expectations of manufacturers reserving the eye-level location, but also maximizes overall category performance. Third, our results provide guidance for structuring slotting fee agreements between retailers and manufacturers, which are often negotiated under information asymmetry and misaligned incentives (Rao and Mahi 2003). By sharing information on the dynamics between eye-level and stretch-level products, both parties can negotiate more efficient terms. For example, if a contract requires placing a product at eye-level despite limited incremental benefit due to a suboptimal planogram reorganization, retailers could justifiably request higher slotting fees to offset the opportunity cost of deviating from an optimal planogram reorganization. Conversely, manufacturers offering products with low substitution potential may rationally pay a premium for eye-level placement, whereas those selling easy-to-substitute products may place lower value on such positioning. Ultimately, slotting fees should not be based solely on whether a product occupies eye-level space, but should also account for the characteristics of adjacent products and the broader planogram configuration, highlighting the need for a menu of contracts with differentiated terms.

Our study has several limitations. First, we focus on products placed on three specific shelves. Our main finding (i.e., the variation in the eye-level effect between two distinct planogram reorganizations) remains robust to this limitation due to the DiD specification, which carefully accounts for unobserved time-invariant heterogeneity. However, our analysis on the overall sales, which aims to provide further practical insights, may be incomplete. Indeed, the products on the shelves not included in our study might also have been influenced by the planogram reorganization. Future research could explore whether our insights on the overall sales across the three treated shelves still hold when considering all the shelves. Second, while our study focuses on fridge displays, retailers also use a variety of shelving units, such as gondolas, checkout counters, promotion stands, and endcaps. Future studies could assess whether the moderating effect of planogram reorganization differs across these display types. Third, our study examines vertical product placement, but shelf layouts also include other dimensions, such as horizontal location and number of facings. Future research could investigate how vertical product placement interacts with horizontal placement or facing adjustments under different planogram reorganizations to influence the eye-level effect. Lastly, we analyzed the level of preference between eye-level and stretch-level products

because the retailer's shelving units were category-specific, ensuring substitutability. However, in some retail environments (e.g., drug stores, large grocery stores), shelving units may accommodate multiple product categories due to their large widths, making substitution between eye-level and stretch-level products less likely. Future research could explore whether planogram reorganization still moderates the eye-level effect in cases where substitution is minimal or nonexistent.

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Appendix A: Details of the Original Field Experiment

The original field experiment was conducted by the author team in partnership with our retail partner. The experiment aimed at identifying the causal effect of different planograms on product sales, with no intention to focus on eye-level placement. This experiment was developed using a treatment-switch approach, where planograms would be systematically switched over time at the treatment stores, while remaining unchanged at the control stores. This design allows us to employ an identification strategy to accurately estimate the causal effect of different planograms on the sales of focal products moved to eye-level. Similar experimental designs have been used in studies on airline pricing (Cohen et al. 2023) and hotel revenue management (Lopez Mateos et al. 2022). Based on the contractual agreements with suppliers, certain products need to be placed in specific store locations. However, because of the requirements of the field experiments, we received permission to temporarily deviate from these arrangements.

The products are displayed in different shelving units, including fridges, gondolas, checkout desks, and promotion stands, which differ in height and number of vertical shelves. The experiment focused on fridges for the following reasons: First, similar fridges are available in other retail settings, such as supermarkets and grocers, hence enhancing the generalizability of our study. Second, unlike other shelving units, such as gondolas and checkout desks, whose heights range from 40" to 60", the height of fridges is 85", allowing us to examine several vertical positions, including the stretch-level.

We aimed to design an experiment within the real-world retail realm while controlling for potential sources of variation across observations. To balance the retailer's request for minimal disruption to daily store operations with the need for a rigorous study, we implemented the experiment illustrated in Figure 3. First, we observed the initial planograms (i.e., P0) across all treatment and control stores between March 21, 2022, and April 16, 2022 (i.e., Round 0). We then replaced these planograms with P5, P1, and P3 in Treatment Stores A, B, and C, respectively, in Round-1 and observed sales between April 18, 2022, and May 28, 2022. The Round-1 data allowed us to confirm that the experiment could proceed using three planograms per treatment store (i.e., P0, P2 and P4), rather than the six (i.e., P0 to P5) that would otherwise be required for a full-factorial design. Subsequently, we implemented these three planograms across the treatment stores in Rounds 2, 3, and 4 between May 30, 2022, and August 6, 2022, as depicted in Figure 3.

This design allowed us to observe each product set across all three vertical locations in the treatment stores. It also enabled us to observe each product set across these vertical locations over time within the treated stores, thereby, controlling for between-store variation across observations from the treated stores. Additionally, by maintaining the original planograms (i.e., P0) unchanged at the control stores throughout the experiment, we were able to account for temporal variation across observations from the treated stores.

		Pre-intervention	Post-intervention				
		Round-0 Mar 21 – Apr 16	Round-1 Apr 18 – May 28	Round-2 May 30 – Jun 18	Round-3 Jun 20 – Jul 09	Round-4 Jul 11 – Aug 06	
Treatment Store A	Stretch level	P0 A	P5 C	P0 A	P2 C	P4 B	
	Eye level	B	B	B	A	C	
	Stoop level	C	A	C	B	A	
Treatment Store B	Stretch level	P0 A	P1 B	P2 C	P4 B	P0 A	
	Eye level	B	A	A	C	B	
	Stoop level	C	C	B	A	C	
Treatment Store C	Stretch level	P0 A	P3 A	P4 B	P0 A	P2 C	
	Eye level	B	C	C	B	A	
	Stoop level	C	B	A	C	B	
Control Stores	Stretch level	P0 A	P0 A	P0 A	P0 A	P0 A	
	Eye level	B	B	B	B	B	
	Stoop level	C	C	C	C	C	

Figure 3 Original field experiment design.

The main experiment began with a baseline period (Round-0), during which we observed the business-as-usual planograms (P0) across the same 14 fridges in treatment stores between March 21, 2022 and April 16, 2022. Following this, we reorganized shelves such that: in Treatment Store 1, products were moved from stretch-level to eye-level creating Planogram 1 (P1), while in Treatment Store 2, products were moved from stoop-level to eye-level, creating Planogram 3 (P3). We then observed sales from April 18, 2022 to May 28, 2022 (Round-1). Next, we maintained the products at eye-level while swapping the products at stretch- and stoop-levels within the treatment stores, resulting in Planogram 2 (P2) in Treatment Store 1 and Planogram 4 (P4) in Treatment Store 2. We observed sales for this phase between May 30, 2022 and June 18, 2022 (Round-2).

In the treatment stores during the experiment, all products on a given shelf were moved vertically together, while their original horizontal locations and number of facings remained unchanged. This selection reflects the retailer's store execution practices, as managers typically prefer to display products from the same brand or category in adjacent locations.

When selecting stores for the treatment and control groups, we considered two factors. First, we identified stores that carry the same set of 14 fridges across all treatment and control stores to ensure consistency in the product assortment between the two groups. Second, the retailer has both traditional stand-alone C-stores and C-stores located in gas stations. Because the customers of these two types of stores are likely to differ in terms of shopping behavior, we limit our study to the traditional stand-alone C-stores. These

two criteria resulted in 20 stores available for our study. Among these, we randomly selected six stores and randomly assigned three of them to the treatment group and three of them to the control group. All six stores are located in the same city (i.e., the largest city in the metropolitan area the retailer serves), ensuring the treated and control stores are comparable in terms of demographic factors related to geographic location (e.g., unemployment rate, household income, population density). Additionally, the treatment and control stores are statistically comparable in size to the broader set of stand-alone C-stores.

To deploy the field experiment, we collaborated with the marketing managers and store staff (i.e., store managers and cashiers). In addition, we formed a field team of five trained research assistants, each assigned to one of the five stores throughout the experiment. At the beginning of the study, before Round-0, the field team visited all five stores to ensure that the business-as-usual planograms (i.e., P0) are properly respected. All planogram implementations at the treated stores were scheduled on Sundays and handled by the field team alongside the store staff. We ensured compliance in two ways. First, the field team visited each store on a daily basis to monitor each planogram, take planogram pictures, ensure that the planograms are organized as planned, and record the inventory, as well as the number of facings per product on each treated shelf. Second, the marketing managers used the same assortment, pricing, and marketing decisions (e.g., promotions, advertising campaigns, etc.) for products in the fridges between the treated and control stores throughout our field experiment.

Our research question in this paper centers on eye-level placement under two different planogram reorganizations. Although the original field experiment was not designed with this specific question in mind, as shown in Figure 3, Treatment Store-B provides a setting where product set A—initially placed at the stretch-level—is moved to eye-level under two different planogram reorganizations in Rounds 1 and 2. Similarly, Treatment Store-C provides a setting where product set C—initially placed at the stoop-level—is moved to eye-level under two different planogram reorganizations in the same rounds. Accordingly, in addition to the three control stores, we focus our analysis on these two treatment stores, which we refer to as Treatment Store 1 (originally Store-B) and Treatment Store 2 (originally Store C). We exclude Treatment Store A because, in the original experiment, it did not implement eye-level placement through two different reorganizations.

Appendix B: Operationalization of HHI

We use pre-intervention data to operationalize HHI for each of the 70 fridges across five stores. For control stores in all three rounds and treatment stores in Round-0, HHI for fridge k in store j is defined as $HHI_{jk} = \sum_{m=1}^{n_{jk}} s_m^2$, where n_{jk} denotes the number of stretch-level and eye-level products in fridge k at store j and s_m represents the percentage share of sales for product m among those products. Hence, in these cases, HHI_{jk} reflects the level of sales concentration within the choice set $\{A,B\}$, where the first letter represents stretch-level products and the second letter represents eye-level products, as illustrated in Figure 2.

To determine HHI for Round-1 and Round-2 in treatment stores, we first estimate the model specified in Equation (6) for each fridge separately. This process results in two fridge-specific coefficients: β_{1jk} represents the effect of stretch-level placement on sales compared to stoop-level placement in fridge k at store j and β_{2jk} denotes the effect of eye-level placement on sales compared to stoop-level placement in fridge k at store j . Using these coefficients and following the choice sets of partial visual searchers in Round-1 and Round-2, we adjust pre-intervention sales based on estimated vertical location effects (β_{1jk} and β_{2jk}) and then calculate HHI using those adjusted sales.

For Round-1 at Treatment Store 1, adjusted sales for the choice set {B,A} are obtained by multiplying pre-intervention sales of products in set B by $(1 + \beta_{1jk})/(1 + \beta_{2jk})$ and those of products in set A by $(1 + \beta_{2jk})/(1 + \beta_{1jk})$. For Round-2 at Treatment Store 1, adjusted sales for the choice set {C,A} are calculated by multiplying pre-intervention sales of products in set C by $(1 + \beta_{1jk})$ and those of products in set A by $(1 + \beta_{2jk})/(1 + \beta_{1jk})$.

Similarly, for Treatment Store 2, adjusted sales for the choice set {A,C} in Round-1 are obtained by multiplying pre-intervention sales of products in set C by $(1 + \beta_{2jk})$, whereas sales of products in set A remain the same. Lastly, adjusted sales for the choice set {B,C} in Round-2 are calculated by multiplying pre-intervention sales of products in set B with $(1 + \beta_{1jk})/(1 + \beta_{2jk})$ and those of products in set C with $(1 + \beta_{2jk})$.

By operationalizing HHI as a time-invariant variable through this method, we can assess whether the changes in HHI expected to occur post-intervention predict post-intervention sales outcomes.

Appendix C: Counterfactual Analysis

To conduct the counterfactual analysis, we first estimate the random slope model in Equation (9), using the natural logarithm of the average number of items sold per facing in fridge k at store j on day t the dependent variable. We then generate out-of-sample predictions under three reorganization scenarios as follows:

- We set $Reorg_1$ to 1 and $Reorg_2$ to 0.
- For the first scenario (i.e., random reorganization), we randomly generate a fridge-level variable to determine whether stretch-level products in Treatment Store 1 or stoop-level products in Treatment Store 2 are moved to eye-level using either planogram reorganization. For Reorganization-1, *Intervention11* and *Intervention12* for Treatment Store 1 observations (and *Intervention21* and *Intervention22* for Treatment Store 2 observations) are set to 1 and 0, respectively. For Reorganization-2, *Intervention11* and *Intervention12* for Treatment Store 1 observations (and *Intervention21* and *Intervention22* for Treatment Store 2 observations) are set to 0 and 1, respectively. We then compute predicted sales ($\hat{Sales}_{ijkt}^1$) using the estimated coefficients from Equation (9) with these updated values. This prediction

reflects expected sales if the retailer moved stretch-level products in Treatment Store 1 and stoop-level products in Treatment Store 2 to eye-level via random planogram reorganization.

- For the second scenario (i.e., uniform reorganization), we set *Intervention11* and *Intervention12* for Treatment Store 1 observations (and *Intervention21* and *Intervention22* for Treatment Store 2 observations) to 1 and 0, respectively, and compute predicted sales ($\hat{Sales}_{ijkt}^2$) using the same estimated coefficients with these updated values. This prediction represents expected sales if the retailer moved stretch-level products in Treatment Store 1 and stoop-level products in Treatment Store 2 to eye-level using Reorganization-1.
- For the third scenario (i.e., tailored reorganization), we generate a fridge-level variable to determine whether Reorganization-1 or Reorganization-2 is applied based on the estimated coefficients in Equation (9). Specifically, if $\beta_1 + \xi_{1ijk} > \beta_2 + \xi_{2ijk}$ for a fridge in Treatment Store 1 (or $\beta_3 + \xi_{3ijk} > \beta_4 + \xi_{4ijk}$ for a fridge in Treatment Store 2), then the retailer follows Reorganization-1, otherwise, Reorganization-2 is applied. We then compute predicted sales ($\hat{Sales}_{ijkt}^3$) following the same approach as in the first scenario. This prediction captures expected sales if the retailer moved stretch-level products in Treatment Store 1 and stoop-level products in Treatment Store 2 to eye-level using a reorganization strategy tailored to each fridge.

Thus, the difference in mean predicted sales between (i) $\hat{Sales}_{ijkt}^2$ and $\hat{Sales}_{ijkt}^1$ represents the counterfactual impact of uniform vs. random reorganization and (ii) $\hat{Sales}_{ijkt}^3$ and $\hat{Sales}_{ijkt}^1$ represents the counterfactual impact of tailored vs. random reorganization.